Pell equations

L4-X4 Level 4 Olympiad · Number theory · Beyond the textbook

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The equation x^2 - dy^2 = 1 and its relatives: the fundamental solution, the descent that generates every solution, the negative Pell equation, classes of solutions of x^2 - dy^2 = n, and how contest problems hide them.
By the end you can find the fundamental solution of a Pell equation and prove that its powers give every solution; solve x^2 - dy^2 = n completely by finding its classes of solutions; recognise a Pell equation hidden in a contest problem and handle the side conditions.
Prerequisites: Olympiad Diophantine equations (L4-NT2). **Contests:** BMO Round 1, BMO Round 2, AIME, USAJMO, USAMO. **Time:** about 120 minutes.

1 Lesson

Find all positive integers n for which $3n+1$ and $4n+1$ are both perfect squares.

A short search finds $n = 56$, with $3n+1 = 13^2$ and $4n+1 = 15^2$, and nothing else below 10 000. Writing $3n+1 = a^2$ and $4n+1 = b^2$ and eliminating n gives $4a^2 - 3b^2 = 1$. This does not factorise over the integers, nothing bounds a and b, and no modulus rules it out, so the methods of L4-NT2 do not finish the problem. There are infinitely many solutions, and the next one is $n = 10\,920$.

The equation $x^2 - dy^2 = 1$ is the **Pell equation**. Its theory is old: Brahmagupta (628) knew how to combine solutions, Bhaskara II (1150) gave an algorithm for finding one, Fermat challenged the English mathematicians with $x^2 - 61y^2 = 1$ in 1657, and Lagrange proved in 1768 that a solution always exists. Euler attached Pell's name by mistake. School courses skip it because every Diophantine equation met there has finitely many solutions. A contest problem that hides a Pell equation is exactly one where that fails, and the standard methods can find the small solutions but cannot prove that the list is complete.

Throughout, d is a positive integer that is not a perfect square, so $\sqrt{d}$ is irrational. For $u = x + y\sqrt{d}$ with x, y integers we write $\bar{u} = x - y\sqrt{d}$ and $N(u) = u\bar{u} = x^2 - dy^2$, the *norm* of u. Since $\sqrt{d}$ is irrational, u determines x and y.

1.1 Multiplying solutions

For all integers x_1, y_1, x_2, y_2,

$$(x_1^2 - dy_1^2)(x_2^2 - dy_2^2) = (x_1x_2 + dy_1y_2)^2 - d(x_1y_2 + x_2y_1)^2.$$

This is Brahmagupta's identity. In the notation above it says $N(uv) = N(u)N(v)$, because $(x_1 + y_1\sqrt{d})(x_2 + y_2\sqrt{d}) = (x_1x_2 + dy_1y_2) + (x_1y_2 + x_2y_1)\sqrt{d}$ and conjugation respects products. In particular, if $N(u) = 1$ then $\bar{u} = 1/u$, and every power u^k with $k \in \mathbb{Z}$ has integer coordinates and norm 1. So one solution of $x^2 - dy^2 = 1$ with $y \neq 0$ produces infinitely many. For $d = 3$, $(2+\sqrt{3})^2 = 7 + 4\sqrt{3}$ and $(2+\sqrt{3})^3 = 26 + 15\sqrt{3}$, and indeed $26^2 - 3 \cdot 15^2 = 1$.

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1.2 Existence and the fundamental solution

Theorem 1 (Lagrange).
The equation $x^2 - dy^2 = 1$ has a solution in positive integers.

Proof. sketch. For each N, two of the $N+1$ fractional parts $\{k\sqrt{d}\}$, $0 \le k \le N$, lie in the same interval $[\frac{j}{N}, \frac{j+1}{N})$; subtracting gives integers x and $1 \le y \le N$ with $|x - y\sqrt{d}| < \frac{1}{N} \le \frac{1}{y}$. Then $x + y\sqrt{d} < 2y\sqrt{d} + 1$, so $|x^2 - dy^2| < 2\sqrt{d} + 1$. Since $x - y\sqrt{d}$ is never 0, letting N grow gives infinitely many such pairs. By pigeonhole, infinitely many share the same value $k = x^2 - dy^2$ (a non-zero integer) and the same residues of x and y modulo $|k|$. Take two of them, $u_1 = x_1 + y_1\sqrt{d}$ and $u_2 = x_2 + y_2\sqrt{d}$ with $u_1 \neq u_2$. Then $u_1\bar{u}_2 = A + B\sqrt{d}$ with $A = x_1x_2 - dy_1y_2 \equiv x_1^2 - dy_1^2 \equiv 0$ and $B = x_2y_1 - x_1y_2 \equiv 0 \pmod{|k|}$, and $N(u_1\bar{u}_2) = k^2$. So $(A/k)^2 - d(B/k)^2 = 1$ in integers. If $B = 0$, then $u_1 = \frac{A}{k}u_2$ and comparing norms gives $\frac{A}{k} = \pm 1$, which is impossible as $u_1 \neq u_2$ and both are positive. So $(|A/k|, |B/k|)$ is a positive solution. □

Among the positive solutions, the one with the smallest y is the **fundamental solution** (x_1, y_1), and we write $\varepsilon = x_1 + y_1\sqrt{d}$. For positive solutions $x = \sqrt{1 + dy^2}$ increases with y, so ε is also the smallest number greater than 1 of the form $x + y\sqrt{d}$ with x, y positive and norm 1. For small d, try $y = 1, 2, 3, \ldots$ until $dy^2 + 1$ is a square: $d = 2, 3, 5, 6, 7, 8$ give $(3,2)$, $(2,1)$, $(9,4)$, $(5,2)$, $(8,3)$, $(3,1)$. Near a square there are formulas: $d = m^2 - 1$ has fundamental solution $(m, 1)$, and $d = m^2 \pm 2$ has the solution $(m^2 \pm 1, m)$. Elsewhere the fundamental solution can be huge: for $d = 13$ it is $(649, 180)$ and for $d = 61$ it is $(1766319049, 226153980)$. Continued fractions find it in general; contest problems choose d so that a search finds it.

1.3 All solutions: the descent

Theorem 2.
The positive solutions of $x^2 - dy^2 = 1$ are exactly the pairs (x_k, y_k), $k \ge 1$, given by $x_k + y_k\sqrt{d} = \varepsilon^k$. All integer solutions are given by $x + y\sqrt{d} = \pm\varepsilon^k$ with $k \in \mathbb{Z}$.

Proof. Let (x, y) be a positive solution and $u = x + y\sqrt{d}$. Then $u \ge \varepsilon > 1$, so there is $k \ge 1$ with $\varepsilon^k \le u < \varepsilon^{k+1}$. Put $v = u\varepsilon^{-k} = u\bar{\varepsilon}^k$. Since $\bar{\varepsilon} = x_1 - y_1\sqrt{d}$ has integer coordinates, $v = a + b\sqrt{d}$ with a, b integers; $N(v) = N(u)N(\bar{\varepsilon})^k = 1$; and $1 \le v < \varepsilon$. Suppose $v > 1$. Then $\bar{v} = 1/v$ lies in $(0, 1)$, so $a = \frac{v+\bar{v}}{2} > 0$ and $b\sqrt{d} = \frac{v-\bar{v}}{2} > 0$. So (a, b) is a positive solution with $a + b\sqrt{d} < \varepsilon$, contradicting the minimality of ε. Hence $v = 1$ and $u = \varepsilon^k$. For an arbitrary solution, $x + y\sqrt{d}$ is one of $\pm u, \pm\bar{u} = \pm u^{-1}$ for a positive solution u, or ± 1 (when $y = 0$). □

This is a descent in the sense of L4-NT2: dividing by ε turns a solution into a smaller one, and the process can only stop at 1. A marker wants to see the three facts that make it work: v has integer coordinates, v has norm 1, and $1 \le v < \varepsilon$ forces positive coordinates.

Comparing coefficients in $\varepsilon^{k+1} = \varepsilon^k\varepsilon$ gives $x_{k+1} = x_1x_k + dy_1y_k$ and $y_{k+1} = y_1x_k + x_1y_k$. Also $\varepsilon + \bar{\varepsilon} = 2x_1$ and $\varepsilon\bar{\varepsilon} = 1$, so both sequences satisfy $s_{k+1} = 2x_1s_k - s_{k-1}$, and $x_k = \frac{\varepsilon^k + \bar{\varepsilon}^k}{2}$, $y_k = \frac{\varepsilon^k - \bar{\varepsilon}^k}{2\sqrt{d}}$. Because these recurrences can be run backwards, the solutions are periodic modulo any m, which is how parity and divisibility side conditions are handled.

Example 1.
Find all positive integers n for which $3n + 1$ and $4n + 1$ are both perfect squares.

Solution.

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Write $3n+1=a^2$ and $4n+1=b^2$ with a,b positive. Then $4a^2-3b^2=1$, that is $X^2-3b^2=1$ with $X=2a$. Conversely, if $X^2-3b^2=1$ with X even, then $3b^2=X^2-1\equiv 3 \pmod 4$, so b is odd, $n=\frac{b^2-1}{4}$ is an integer, $4n+1=b^2$ and $3n+1=\frac{3b^2+1}{4}=a^2$. So we need the solutions of $X^2-3Y^2=1$ with X even.

The fundamental solution is $(2,1)$, so by Theorem 2 the positive solutions are $X_k+Y_k\sqrt{3}=(2+\sqrt{3})^k$, with $X_{k+1}=2X_k+3Y_k$ and $Y_{k+1}=X_k+2Y_k$. Modulo 2 this reads $(X_{k+1},Y_{k+1})\equiv(Y_k,X_k)$, and $(X_1,Y_1)\equiv(0,1)$, so X_k is even exactly when k is odd. The odd powers are $(2,1)$, $(26,15)$, $(362,209)$, $(5042,2911)$, and so on; the first gives $n=0$. Hence the answers are $n=\frac{Y^2-1}{4}$ for $Y=15,209,2911,\ldots$, where Y runs through the sequence $1,15,209,\ldots$ with $Y_{j+1}=14Y_j-Y_{j-1}$ (the trace of $(2+\sqrt{3})^2=7+4\sqrt{3}$ is 14). The first three values are $n=56$, 10 920 and 2 118 480.

Remark. The search finds 56, and the factorisation and modular methods of L4-NT2 say nothing about completeness. Theorem 2 is what proves that there are no other solutions.

1.4 The negative Pell equation

The equation $x^2-dy^2=-1$ may or may not have solutions. If it does, then $x^2\equiv -1$ modulo every prime factor of d and modulo 4 if $4\mid d$, so $4\nmid d$ and every odd prime factor of d is 1 mod 4. The converse fails: $d=34$ passes this test and has no solutions (Practice 5).

Theorem 3.
Let η be the smallest number greater than 1 of the form $x+y\sqrt{d}$ with x,y integers and $x^2-dy^2=\pm 1$. Every such number that is positive is η^k for some $k\in\mathbb{Z}$. If $N(\eta)=1$, then $\eta=\varepsilon$ and $x^2-dy^2=-1$ has no solutions. If $N(\eta)=-1$, then $\varepsilon=\eta^2$, and the positive solutions of $x^2-dy^2=-1$ are given by the odd powers $\eta,\eta^3,\eta^5,\ldots$

Proof. First, η exists: a number $u=x+y\sqrt{d}>1$ with $N(u)=\pm 1$ has $|\bar{u}|=1/u<1$, so $x=\frac{u+\bar{u}}{2}>0$ and $y\sqrt{d}=\frac{u-\bar{u}}{2}>0$, and there are finitely many such u below any bound. Now repeat the proof of Theorem 2 with η in place of ε: for such a u, choose k with $\eta^k\le u<\eta^{k+1}$; then $v=u\eta^{-k}$ has integer coordinates, norm ± 1 and $1\le v<\eta$, and the argument just given shows that $v>1$ would make v a smaller candidate than η. So $u=\eta^k$, and $N(\eta^k)=N(\eta)^k$ gives the rest. □

For $d=2$, $\eta=1+\sqrt{2}$, so the solutions of $x^2-2y^2=-1$ are $(1,1)$, $(7,5)$, $(41,29)$, …, and $\varepsilon=(1+\sqrt{2})^2=3+2\sqrt{2}$. For $d=m^2+1$, $\eta=m+\sqrt{d}$.

1.5 The general equation $x^2-dy^2=n$

Multiplying a solution of $x^2-dy^2=n$ by ε gives another, so solutions come in *classes* $\{v\varepsilon^k : k\in\mathbb{Z}\}$. There may be several classes, or none, and the theorem below finds them all by a finite search.

Theorem 4.
Let $n\neq 0$. Every $u=x+y\sqrt{d}$ with x,y integers, $N(u)=n$ and $u>0$ can be written as $u=v\varepsilon^k$ with $k\in\mathbb{Z}$ and $v=a+b\sqrt{d}$ a solution with $\sqrt{|n|}\le v<\varepsilon\sqrt{|n|}$. Every such v satisfies $|a|<\frac{(\varepsilon+1)\sqrt{|n|}}{2}$ and $|b|<\frac{(\varepsilon+1)\sqrt{|n|}}{2\sqrt{d}}$. Moreover x and y are both positive if and only if $u>\sqrt{|n|}$.

Proof. Choose k with $\varepsilon^k\sqrt{|n|}\le u<\varepsilon^{k+1}\sqrt{|n|}$ and put $v=u\varepsilon^{-k}$; as before v has integer coordinates

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and $N(v) = n$. From $|\bar{v}| = |n|/v \le \sqrt{|n|}$ we get $|a| = \frac{|v+\bar{v}|}{2}$ and $|b|\sqrt{d} = \frac{|v-\bar{v}|}{2}$, both less than $\frac{\varepsilon\sqrt{|n|}+\sqrt{|n|}}{2}$. Finally, $x = \frac{u+\bar{u}}{2}$ and $y\sqrt{d} = \frac{u-\bar{u}}{2}$ are both positive if and only if $u > |\bar{u}| = |n|/u$, that is $u > \sqrt{|n|}$. □

So the positive solutions are the numbers $v\varepsilon^k$ with $k \ge 0$ and v in the window, except v itself when $v = \sqrt{|n|}$.

Example 2.
Find all positive integers y for which $2y^2 + 7$ is a perfect square.

Solution.
We need the positive solutions of $x^2 - 2y^2 = 7$. Here $\varepsilon = 3 + 2\sqrt{2} \approx 5.83$, and the window is $\sqrt{7} \le a + b\sqrt{2} < (3 + 2\sqrt{2})\sqrt{7}$, roughly $[2.65, 15.42)$. Theorem 4 gives $|a| \le 9$ and $|b| \le 6$. Checking $b = 0, \pm1, \ldots, \pm6$, the only solutions of $a^2 = 2b^2 + 7$ in this range are $(a, b) = (\pm3, \pm1)$ and $(\pm5, \pm3)$, and of the eight numbers $a + b\sqrt{2}$ only $3 + \sqrt{2} \approx 4.41$ and $5 + 3\sqrt{2} \approx 9.24$ lie in the window. So there are two classes, and the positive solutions are $(3 + \sqrt{2})\varepsilon^k$ and $(5 + 3\sqrt{2})\varepsilon^k$, $k \ge 0$:

$$3 + \sqrt{2},\ 13 + 9\sqrt{2},\ 75 + 53\sqrt{2},\ \ldots \qquad 5 + 3\sqrt{2},\ 27 + 19\sqrt{2},\ 157 + 111\sqrt{2},\ \ldots$$

The values of y are $1, 9, 53, 309, \ldots$ and $3, 19, 111, \ldots$, each sequence satisfying $y_{k+1} = 6y_k - y_{k-1}$.
Remark. The two classes are conjugate: $5 + 3\sqrt{2} = (3 - \sqrt{2})\varepsilon$. Students often find the first class and stop. Without the window argument there is no proof that the list is complete.

1.6 Pell equations in disguise

Any equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ whose discriminant $B^2 - 4AC$ is positive and not a square becomes $X^2 - dY^2 = n$ after completing the square, with X and Y linear in x and y. The cost is a congruence condition on X and Y (parity, or a residue modulo $2A$), which is handled with the periodicity noted after Theorem 2. Vieta jumping (replacing one root of a quadratic in a by the other root) is the same descent: after completing the square it is multiplication by a unit of norm 1.

Example 3.
In a room of a people there are three times as many handshakes as in a room of b people, where $a, b \ge 2$ and every pair of people in a room shakes hands once. Find all possible pairs (a, b).

Solution.
The condition is $a(a-1) = 3b(b-1)$. Since a and $a-1$ are coprime, one could split the factors of $3b(b-1)$ between them, but that produces several cases, each an equation of the same type as the original. Instead complete the square: $4a^2 - 4a = 12b^2 - 12b$ gives $(2a-1)^2 - 1 = 3\big((2b-1)^2 - 1\big)$, that is

$$X^2 - 3Y^2 = -2, \qquad X = 2a - 1,\ Y = 2b - 1.$$

Conversely every solution with X, Y positive and odd gives a pair. Every solution of $X^2 - 3Y^2 = -2$ has X and Y odd: if both were even the left side would be divisible by 4, and if exactly one were even it would be odd. So we need all positive solutions. With $\varepsilon = 2 + \sqrt{3}$ the window is $[\sqrt{2}, (2 + \sqrt{3})\sqrt{2}) \approx [1.41, 5.28)$, and Theorem 4 gives $|X| \le 3$, $|Y| \le 1$. The

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solutions there are $(\pm 1, \pm 1)$, and only $1+\sqrt{3}$ lies in the window. So there is one class, and the positive solutions are $(1+\sqrt{3})(2+\sqrt{3})^k$, $k \geq 0$:

$$1+\sqrt{3},\ 5+3\sqrt{3},\ 19+11\sqrt{3},\ 71+41\sqrt{3},\ 265+153\sqrt{3},\ \ldots$$

The first gives $a = b = 1$, which is excluded. The rest give $(a,b) = (3,2), (10,6), (36,21), (133,77), \ldots$; both a and b satisfy $s_{k+1} = 4s_k - s_{k-1} - 1$, because X and Y satisfy $s_{k+1} = 4s_k - s_{k-1}$. For instance $\binom{36}{2} = 630 = 3\binom{21}{2}$.
Remark. Vieta jumping is no help here: as a quadratic in a, the other root of $a^2 - a - 3b(b-1) = 0$ is $1-a$, and similarly for b, because the equation has no ab term. Completing the square supplies the unit $2+\sqrt{3}$, which moves both variables at once.

Common traps.

- If d is a perfect square, $x^2 - dy^2$ factorises and there are finitely many solutions; none of the theory applies. Check that d is not a square after completing the square.
- Finding the solutions is not solving the problem. A complete answer proves that the list is complete, using Theorem 2 for $n = \pm 1$ and the window of Theorem 4 for other n.
- For $n \neq \pm 1$ there can be several classes (Example 2). Check every v in the window, including those with negative coordinates.
- Side conditions (parity, divisibility by 3, $X = 2a - 1$) select a subsequence. Prove which one by periodicity of the recurrence, not by looking at three terms.
- The necessary conditions for $x^2 - dy^2 = -1$ are not sufficient, and $d = 34$ is the standard counterexample.
- Do not forget the trivial solutions $(\pm 1, 0)$, the negative solutions, and whether the problem asks for positive or non-negative values.

1.7 How to spot this in a contest

1. Two quadratic (or linear) expressions in one variable n must both be squares: eliminate n.
2. Two figurate numbers are equal, or one is a fixed multiple of another: triangular numbers, $\binom{a}{2}$ and $\binom{b}{2}$, squares, sums of consecutive squares or cubes.
3. A quadratic equation in two variables whose discriminant is positive and not a square: complete the square.
4. The problem asks you to prove that infinitely many integers have a quadratic property, or to find the largest or the k-th solution.
5. A sequence with $u_{k+1} = cu_k - u_{k-1}$, or with terms such as $\sqrt{5y^2 - 4}$, or a pair (a,b) with $ab \mid a^2 + b^2 + c$: this is descent on a Pell-type equation.

Once the equation is in the form $X^2 - dY^2 = n$, the work is always the same: find ε, find the classes with Theorem 4, then impose the side conditions with the recurrence.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Find the fundamental solution of $x^2 - 11y^2 = 1$, and the next two solutions in positive integers.

Problem 2 ⋆⋆ Find the four smallest positive integers n for which $n+1$ and $5n+1$ are both perfect squares.

Problem 3 ⋆⋆ Prove that there are infinitely many positive integers n such that $n^2 + n - 1$ is five times a perfect square.

Problem 4 ⋆⋆⋆ Find all positive integers y less than 1000 for which $3y^2 + 13$ is a perfect square.

Problem 5 ⋆⋆⋆ Prove that the equation $x^2 - 34y^2 = -1$ has no solutions in integers, even though the congruence $x^2 \equiv -1 \pmod{34}$ has solutions.

Problem 6 ⋆⋆⋆⋆ Let (x_k, y_k), $k = 1, 2, 3, \ldots$, be the solutions of $x^2 - 2y^2 = 1$ in positive integers, in increasing order, so $(x_1, y_1) = (3, 2)$ and $(x_2, y_2) = (17, 12)$. Prove that $\gcd(y_m, y_n) = y_{\gcd(m,n)}$ for all $m, n \geq 1$, and find the smallest n for which 200 divides y_n.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME II 2008 Problem 15** ⋆⋆
   A difference of consecutive cubes equal to a square is a Pell equation after completing the square; the second condition picks out the solutions.

2. **HMMT February 2017 Guts Round Problem 27** ⋆⋆
   Searching for the fundamental solution directly is slow; 29 is close to a square, so look at the negative equation first.

3. **BMO2 2017/18 Problem 3** ⋆⋆
   Find the sum in closed form; a quadratic factor must be a square. Complete the square and study the resulting Pell-type equation modulo a small number.

4. **HMMT February 2020 Guts Round Problem 21** ⋆⋆
   Show that a quadratic expression in consecutive terms is invariant, so the sequence obeys a linear recurrence of the form a(n+1) = c a(n) - a(n-1).

5. **BMO2 2001/02 Problem 3** ⋆⋆⋆
   Show that $5y^2$ - 4 is a square for every term, and write the step from one term to the next as multiplication by a unit.

6. **IMO 1981 Problem 3** ⋆⋆⋆
   Show that one solution produces a smaller one, and descend.

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7. **HMMT February 2017 Algebra and Number Theory Problem 8** ⋆⋆⋆
   Fix the value of the quotient and jump between the two roots of the quadratic to find all solutions from the smallest ones.

8. **BMO2 2006/07 Problem 2** ⋆⋆⋆
   Clear denominators and treat the equation as a quadratic in one variable: the other root gives a new solution.

9. **IMO 1988 Problem 6** ⋆⋆⋆⋆
   Fix the quotient and descend from a minimal solution using the other root of the quadratic.

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4 Hints

1. Try $y = 1, 2, 3$ in turn. Then square and cube $x_1 + y_1\sqrt{11}$.
2. Eliminate n to get $b^2 - 5a^2 = -4$, and find its classes with Theorem 4.
3. Multiply by 4 and complete the square. You need one solution and a way to make more that keeps the parity right.
4. Find all classes of $x^2 - 3y^2 = 13$ with Theorem 4. There is more than one.
5. Find the fundamental solution of $x^2 - 34y^2 = 1$. If the negative equation had a solution, Theorem 3 would make ε a square.
6. Put $(x_0, y_0) = (1, 0)$ and use $\varepsilon^m = \varepsilon^{m-n}\varepsilon^n$ to run the Euclidean algorithm on the indices. For the second part, find the indices n with $8 \mid y_n$ and with $25 \mid y_n$ separately.

5 Solutions

Problem 1

Find the fundamental solution of $x^2 - 11y^2 = 1$, and the next two solutions in positive integers.

For $y = 1$ and $y = 2$, $11y^2 + 1$ is 12 and 45, neither a square; for $y = 3$ it is $100 = 10^2$. So the fundamental solution is $(10, 3)$ and $\varepsilon = 10 + 3\sqrt{11}$. Then $\varepsilon^2 = 100 + 60\sqrt{11} + 99 = 199 + 60\sqrt{11}$, and $\varepsilon^3 = (199 + 60\sqrt{11})(10 + 3\sqrt{11}) = 1990 + 1980 + (597 + 600)\sqrt{11} = 3970 + 1197\sqrt{11}$. By Theorem 2 these are the next two solutions: $(199, 60)$ and $(3970, 1197)$. As a check, $199^2 - 11 \cdot 60^2 = 39601 - 39600 = 1$. (Note that $11 = 3^2 + 2$, so the formula $(m^2 + 1, m)$ with $m = 3$ also gives $(10, 3)$.)

Answer: $(10, 3)$, then $(199, 60)$ and $(3970, 1197)$

Problem 2

Find the four smallest positive integers n for which $n + 1$ and $5n + 1$ are both perfect squares.

Write $n + 1 = a^2$ and $5n + 1 = b^2$ with a, b positive. Then $b^2 - 5a^2 = -4$, and conversely every positive solution with $a \geq 2$ gives $n = a^2 - 1 \geq 1$ with $5n + 1 = 5a^2 - 4 = b^2$. The fundamental solution of $x^2 - 5y^2 = 1$ is $(9, 4)$, so $\varepsilon = 9 + 4\sqrt{5} \approx 17.94$. By Theorem 4 every positive solution of $b^2 - 5a^2 = -4$ is $v\varepsilon^k$ with $k \geq 0$ and $v = b' + a'\sqrt{5}$ in the window $[2, 2\varepsilon) \approx [2, 35.9)$, where $|b'| \leq 18$ and $|a'| \leq 8$. Checking $a' = 0, \pm 1, \ldots, \pm 8$ for which $5a'^2 - 4$ is a square gives $(b', a') = (\pm 1, \pm 1)$, $(\pm 4, \pm 2)$, $(\pm 11, \pm 5)$, and the ones in the window are $1 + \sqrt{5} \approx 3.24$, $4 + 2\sqrt{5} \approx 8.47$ and $11 + 5\sqrt{5} \approx 22.2$.

So the positive solutions are these three times ε^k, $k \geq 0$. Multiplying by ε gives $29 + 13\sqrt{5}$, $76 + 34\sqrt{5}$ and $199 + 89\sqrt{5}$, and further powers give $a > 89$. So the values of a in increasing order are $1, 2, 5, 13, 34, 89, \ldots$ (every other Fibonacci number). Discarding $a = 1$, which gives $n = 0$, the four smallest values are $n = 2^2 - 1 = 3$, $5^2 - 1 = 24$, $13^2 - 1 = 168$ and $34^2 - 1 = 1155$. For example $5 \cdot 1155 + 1 = 5776 = 76^2$.

Answer: 3, 24, 168, 1155

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Problem 3

Prove that there are infinitely many positive integers n such that n^2+n-1 is five times a perfect square.

We have $n^2+n-1=5m^2$ if and only if $4n^2+4n-4=20m^2$, that is $(2n+1)^2-20m^2=5$. So it suffices to find infinitely many solutions of $X^2-20m^2=5$ in positive integers with X odd and $X\geq 3$, since then $n=\frac{X-1}{2}$ is a positive integer with $n^2+n-1=5m^2$.

The pair $(X,m)=(5,1)$ is a solution, and $(9,2)$ solves $x^2-20y^2=1$ because $81-80=1$. Define $(X_0,m_0)=(5,1)$ and $X_{k+1}+m_{k+1}\sqrt{20}=(X_k+m_k\sqrt{20})(9+2\sqrt{20})$, that is

$$X_{k+1}=9X_k+40m_k, \qquad m_{k+1}=2X_k+9m_k.$$

By Brahmagupta's identity each pair satisfies $X_k^2-20m_k^2=5\cdot 1=5$. By induction X_k and m_k are positive and $X_{k+1}>X_k$, so the pairs are distinct. Also $X_{k+1}\equiv 9X_k\equiv X_k \pmod 2$, so every X_k is odd. Hence $n_k=\frac{X_k-1}{2}$ gives infinitely many positive integers with the required property. The first three are $n=2, 42$ and 762; for instance $42^2+42-1=1805=5\cdot 19^2$.

Problem 4

Find all positive integers y less than 1000 for which $3y^2+13$ is a perfect square.

We need the positive solutions of $x^2-3y^2=13$. Here $\varepsilon=2+\sqrt{3}$, and by Theorem 4 every positive solution is $v\varepsilon^k$ with $k\geq 0$ and $v=a+b\sqrt{3}$ a solution with $\sqrt{13}\leq v<(2+\sqrt{3})\sqrt{13}$, roughly $[3.61, 13.46)$, and $|a|<\frac{(3+\sqrt{3})\sqrt{13}}{2}\approx 8.5$, $|b|<4.93$. For $b=0,\pm1,\pm2,\pm3,\pm4$, the number $3b^2+13$ is $13, 16, 25, 40, 61$, so the solutions in range are $(\pm4,\pm1)$ and $(\pm5,\pm2)$. Of these, only $4+\sqrt{3}\approx 5.73$ and $5+2\sqrt{3}\approx 8.46$ lie in the window ($4-\sqrt{3}$ and $5-2\sqrt{3}$ are below 3.61, and the others are negative). So there are two classes.

Multiplying repeatedly by $2+\sqrt{3}$:

$$4+\sqrt{3},\ 11+6\sqrt{3},\ 40+23\sqrt{3},\ 149+86\sqrt{3},\ 556+321\sqrt{3},\ 2075+1198\sqrt{3},$$

$$5+2\sqrt{3},\ 16+9\sqrt{3},\ 59+34\sqrt{3},\ 220+127\sqrt{3},\ 821+474\sqrt{3},\ 3064+1769\sqrt{3}.$$

The y-coordinates increase along each class, so the values below 1000 are $1, 6, 23, 86, 321$ and $2, 9, 34, 127, 474$. In increasing order: $1, 2, 6, 9, 23, 34, 86, 127, 321, 474$, ten values. For example $3\cdot 474^2+13=674041=821^2$.

Answer: $1, 2, 6, 9, 23, 34, 86, 127, 321, 474$

Problem 5

Prove that the equation $x^2-34y^2=-1$ has no solutions in integers, even though the congruence $x^2\equiv -1 \pmod{34}$ has solutions.

First, $13^2=169=5\cdot 34-1$, so $x^2\equiv -1 \pmod{34}$ is solvable, and no congruence condition of the kind in the lesson rules the equation out.

For $y=1,2,3,4,5$ the number $34y^2+1$ is $35, 137, 307, 545, 851$, none a square (they lie between consecutive squares: $5^2<35<6^2$, $11^2<137<12^2$, $17^2<307<18^2$, $23^2<545<24^2$, $29^2<851<30^2$), and for $y=6$ it is $1225=35^2$. So the fundamental solution of $x^2-34y^2=1$ is $(35,6)$ and $\varepsilon=35+6\sqrt{34}$.

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Suppose $x^2 - 34y^2 = -1$ has a solution. Then by Theorem 3 the number η defined there has norm -1, and $\eta^2 = \varepsilon$. Also $\eta > 1$, so $\eta = a + b\sqrt{34}$ with a, b positive integers (shown in the proof of Theorem 3). Comparing coefficients in $(a + b\sqrt{34})^2 = 35 + 6\sqrt{34}$ gives $2ab = 6$ and $a^2 + 34b^2 = 35$. From $ab = 3$ we have $(a, b) = (1, 3)$ or $(3, 1)$, giving $a^2 + 34b^2 = 307$ or 43, not 35. This contradiction shows that there are no solutions.

Problem 6

Let (x_k, y_k), $k = 1, 2, 3, \ldots$, be the solutions of $x^2 - 2y^2 = 1$ in positive integers, in increasing order, so $(x_1, y_1) = (3, 2)$ and $(x_2, y_2) = (17, 12)$. Prove that $\gcd(y_m, y_n) = y_{\gcd(m,n)}$ for all $m, n \geq 1$, and find the smallest n for which 200 divides y_n.

Let $\varepsilon = 3 + 2\sqrt{2}$, the fundamental solution, so $x_k + y_k\sqrt{2} = \varepsilon^k$ for $k \geq 1$ by Theorem 2; put $(x_0, y_0) = (1, 0)$ so that this also holds for $k = 0$. Since $x_k^2 - 2y_k^2 = 1$, any common divisor of x_k and y_k divides 1, so $\gcd(x_k, y_k) = 1$. For $m \geq n \geq 0$, comparing coefficients of $\sqrt{2}$ in $\varepsilon^m = \varepsilon^{m-n}\varepsilon^n$ gives

$$y_m = x_{m-n}y_n + y_{m-n}x_n.$$

Hence $\gcd(y_m, y_n) = \gcd(y_{m-n}x_n, y_n) = \gcd(y_{m-n}, y_n)$, the last step because $\gcd(x_n, y_n) = 1$. So $\gcd(y_m, y_n)$ is unchanged when the larger index is replaced by the difference of the indices. Running the Euclidean algorithm on (m, n) ends at $(\gcd(m, n), 0)$, so $\gcd(y_m, y_n) = \gcd(y_{\gcd(m,n)}, y_0) = y_{\gcd(m,n)}$, since $y_0 = 0$.

A consequence: for any positive integer M, the set $S_M = \{n \geq 1 : M \mid y_n\}$ is closed under taking gcds, and it contains every multiple of each of its elements because $y_r \mid y_{jr}$ (as $\gcd(y_r, y_{jr}) = y_r$). So if S_M is non-empty, it is the set of multiples of its least element r_M.

The recurrence $y_{k+1} = 2x_k + 3y_k$, $x_{k+1} = 3x_k + 4y_k$ gives $y_1, \ldots, y_4 = 2, 12, 70, 408$. Since $8 \nmid 2, 12, 70$ and $408 = 8 \cdot 51$, we get $r_8 = 4$. Since $5 \nmid 2, 12$ and $5 \mid 70$, $r_5 = 3$, and r_{25} is a multiple of 3 because $S_{25} \subseteq S_5$. Let $z_j = y_{3j}$. As $\varepsilon^3 = 99 + 70\sqrt{2}$ has trace 198 and norm 1, the sequence satisfies $z_{j+1} = 198z_j - z_{j-1}$, with $z_0 = 0$, $z_1 = 70$. Modulo 25, $198 \equiv -2$, and the residues are $z_0, \ldots, z_5 \equiv 0, 20, 10, 10, 20, 0$. So $r_{25} = 15$. Finally $200 \mid y_n$ if and only if $4 \mid n$ and $15 \mid n$, so the smallest such n is 60.

Answer: $n = 60$